

Karthikeyan Devadoss

Forward Deployed Engineer (Physical AI Infrastructure) | Cloud & backend foundation

Berlin, Germany · German permanent residence (Niederlassungserlaubnis) · no visa sponsorship needed · Live AI platform: [agentic-platform-backend-production.up.railway.app](#) · [github.com/karthikdevadoss](#)

SUMMARY

Applying for Forward Deployed Engineer, Physical AI Infrastructure at Nebius. Engineer focused on AI solution architecture, with about ten years of backend development since December 2016. Since early 2026, one of a handful of engineers in an initial core AI team at a large retail-energy company, designing and building AI solutions hands-on alongside senior backend work, including production AI triage with a human approval step, adopted by several teams. 4+ years combining hands-on Java/Spring with architecture discussions and design decisions alongside the lead.

CORE SKILLS

Backend	Java, Spring Boot, REST, GraphQL, JWT/FusionAuth, AWS Lambda/SAM, SQS, Oracle, Docker, Jenkins, JUnit, Kafka, FHIR, Apache Camel
AI architecture	AI solution design · human-approval & capability boundaries · deterministic vs LLM decision split · RAG & retrieval evaluation (recall/MRR) · MCP & agent tool boundaries
AI stack	Models used day-to-day: Claude Sonnet, Claude Code, Grok Bot, Fabel, GPT-4o / Azure OpenAI, Gemini · vector retrieval (production triage) · Python, Anthropic API, embeddings, Postgres, Playwright (personal project)

EXPERIENCE

Senior Backend Developer — NRG Energy

Mar 2022 – present

End client; employed via Remote GmbH · Texas, USA / Berlin

- Java/Spring Boot on a large retail-energy platform serving 5+ million customers; since early 2026, one of a handful of engineers in the initial core AI team — designing and building AI solutions hands-on (production AI triage with Azure OpenAI, vector retrieval, and human email approval before action; adopted by several teams) alongside senior backend work.
- Took part in architecture discussions and design decisions with the lead; REST, GraphQL, JWT/FusionAuth; 2025 AWS Lambda/SAM; mentoring; monthly production on-call.

Java Developer Consultant — Blue Cross Blue Shield Association (BCBSA)

Jun 2019 – Mar 2022

Chicago, USA

- National hub connecting state Blue Cross Blue Shield plans (~118 million members system-wide): Java/Spring Boot REST and FHIR interoperability / payer-to-payer exchange for member, coverage, and claims services.
- Application-side Kafka producers and consumers for domain events; careful handling of sensitive healthcare data.

Java Backend Developer — Marsh (Digital Broker / Bluestream, Affinity)

Dec 2016 – Jun 2019

Via Innovecture LLC · Phoenix, USA · ~95,000 colleagues, 130 countries

- Java/Spring carrier integrations (Apache Camel routes, ACORD-oriented mapping); on IMT-Germany, Data Modeler and System Analyst — built within the technical architect's design.

PERSONAL PROJECT

Agentic Software Delivery Platform

PERSONAL PROJECT

Live demo · GitHub repo

Independently designed and built

- Deterministic vs LLM split:** deterministic checks kept separate from LLM proposals; a Python agent (Anthropic API) plans and applies changes to a real Spring Boot target via tool calling.
- Human-approval boundary:** structural approval gate; the model has no code path to approve its own change, and approval is bound by content hash to the exact change.
- RAG eval design:** local retrieval (embeddings, cosine similarity, content-hash indexing) with a labelled eval (recall/MRR).
- MCP / tool boundaries:** MCP server for read-only repository tools; independent QA subagent without write tools; durable Postgres event ledger with versioned pricing.

EDUCATION

B.E. Computer Science — Madras University, Chennai

LANGUAGES

English — fluent · German — A1-A2